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import math
machines = ['A', 'B', 'C']

M = 3
N_play = 10
```

```
# Number of times each machine is played
N = {machine: 0 for machine in machines}

# Rewards received from each machine
R = {machine: [] for machine in machines}

# Estimated Q-values
Q = {machine: 0.0 for machine in machines}

# UCB values
UCB = {machine: float('inf') for machine in machines}

# Total reward
total_reward = 0
```

```
def calculate_ucb(machine, t):

    # If machine has not been played,
    # give infinite UCB value for exploration
    if N[machine] == 0:
        return float('inf')

    ucb_value = Q[machine] + math.sqrt(
        (2 * math.log(t)) / N[machine]
    )

    return ucb_value
```

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    return ucb_value

for play in range(1, N_play + 1):

    print("\n" + "=" * 40)
    print("Play Number:", play)

    # Calculate UCB for every machine
    for machine in machines:
        UCB[machine] = calculate_ucb(machine, play)

    # Display UCB values
    print("\nUCB Values:")

    for machine in machines:
        if UCB[machine] == float('inf'):
            print(machine, ": Infinity")
        else:
            print(machine, ":", round(UCB[machine], 3))

    # Select machine with highest UCB
    selected_machine = max(UCB, key=UCB.get)

    print("\nPlay with Machine:", selected_machine)

    # Get reward from user
    reward = float(input("Enter reward: "))
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# Store reward
R[selected_machine].append(reward)

# Update number of times selected
N[selected_machine] += 1

# Incremental Sample Average Q-value Update
Q[selected_machine] = Q[selected_machine] + (
    (reward - Q[selected_machine]) / N[selected_machine]
)

# Update total reward
total_reward += reward

print("\nUpdated Q Values:")
for machine in machines:
    print(machine, ":", round(Q[machine], 3))

```

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Play Number: 1

UCB Values:
A : Infinity
B : Infinity
C : Infinity

Play with Machine: A
Enter reward: 3

Updated Q Values:
A : 3.0
B : 0.0
C : 0.0

=====
Play Number: 2

UCB Values:
A : 4.177
B : Infinity
C : Infinity

Play with Machine: B

```

```
print("\n" + "=" * 50)
print("FINAL RESULTS")
print("=" * 50)

print("\nMachine\tN\tRewards\tQ-value\tUCB")

for machine in machines:

    # Calculate final UCB using total plays
    if N[machine] == 0:
        final_ucb = float('inf')
    else:
        final_ucb = Q[machine] + math.sqrt(
            (2 * math.log(N_play)) / N[machine]
        )

    print(
        machine,
        "\t",
        N[machine],
        "\t",
        R[machine],
        "\t",
        round(Q[machine], 3),
        "\t\t",
        round(final_ucb, 3)
    )

print("\nTotal Reward Received:", total_reward)

best_machine = max(Q, key=Q.get)

print("Best Machine:", best_machine)
print("Best Q-value:", round(Q[best_machine], 3))
```